

D - Concat Power of 2

分值: 400 分

Problem Statement

We call a positive integer a **good integer** if it satisfies the following condition.

我们称满足下述条件的正整数为**好整数**。

- Condition: It can be obtained by choosing one or more powers of 2 ($1, 2, 4, 8, 16, \dots$) (repetition and reordering allowed), concatenating them as strings, and interpreting the result as an integer.
- 条件: 选择一个或多个2的幂 (即 $1, 2, 4, 8, 16, \dots$) , 允许重复选择和调整顺序, 将它们按字符串形式拼接后得到的整数即为好整数。

Find the N -th smallest good integer. It is guaranteed that the N -th smallest good integer is at most 10^9 .

求第 N 小的好整数。题目保证第 N 小的好整数不超过 10^9 。

Constraints

- N is a positive integer.
- The N -th smallest good integer is at most 10^9 .
- N 是正整数。
- 第 N 小的好整数不超过 10^9 。

Input

The input is given from Standard Input in the following format:

输入按照以下格式从标准输入给出:

N

Output

Output the answer.

输出答案。

Sample Input 1

10

Sample Output 1

21

Listing good integers in ascending order gives 1, 2, 4, 8, 11, 12, 14, 16, 18, 21,

将所有好整数按升序排列后得到的序列为1, 2, 4, 8, 11, 12, 14, 16, 18, 21,

Sample Input 2

69

Sample Output 2

328

Sample Input 3

1099898

Sample Output 3

819264512